

Experiment 'p_2_2_letter_perimeter' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_2_letter_perimeter

Natural language statement: Write a method that computes and returns the perimeter of a letter-size (8.5 x 11 inches) sheet of paper and the length of its diagonal with a residual error less than 0.01.

Method signature: p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)

Ensures

- `perimeter == (8.5 + 11.0) * 2.0`
- `-0.01 < (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that computes and returns the perimeter of a letter-size (8.5 x 11 inches)
→ sheet of paper and the length of its diagonal with a residual error less than
→ 0.01.

The signature should be:

p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)

The method should respect the following contract:

ensures perimeter == (8.5 + 11.0) * 2.0, ensures -0.01 < (diagonal * diagonal) - ((8.5 *
→ 8.5) + (11.0 * 11.0)) < 0.01

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdWFTIFnFWoGbq4TxzFMU97pzN8X

Model: gpt-5.5-2026-04-23

Created at: 1780484027

Finish reason: stop

Usage: 420 tokens in, and 1677 tokens out

```
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9014;

  assert diagonal * diagonal == 193.24892196;
  assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;
  assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == -0.00107804;
  assert -0.01 < -0.00107804;
  assert -0.00107804 < 0.01;
}
```

Final Program

```
method p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)
  ensures perimeter == (8.5 + 11.0) * 2.0
  ensures -0.01 < (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01
```

```
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9014;

  assert diagonal * diagonal == 193.24892196;
  assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;
  assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == -0.00107804;
  assert -0.01 < -0.00107804;
  assert -0.00107804 < 0.01;
}
```

Total Token Usage

Input tokens: 420

Output tokens: 1677

Reasoning tokens: 1501

Sum of 'total tokens': 2097

Experiment Timings

Overall Experiment started at 1780484027517, ended at 1780484063445, lasting 35928ms (35.93 seconds)

Iteration #1 started at 1780484027518, ended at 1780484063444, lasting 35926ms (35.93 seconds)